CARNATOR: FAST TEXT2VIDEO GENERATION WITH GENERATION-NATIVE COMPATIBILITY-GUIDED CROSS-REQUEST REUSE

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ABSTRACT

Video diffusion transformers produce high-quality videos at substantial computational cost, posing a major challenge for interactive and large-scale serving. Most existing acceleration methods focus on individual requests, limiting their efficiency gains to redundancy within a single generation trajectory. Recent cross-request approaches seek additional savings by reusing computation from related historical generations, but typically rely on coarse semantic similarity, making it difficult to determine whether and where reuse remains valid, risking either quality degradation or limited acceleration. We present *Carnator*, a cross-request acceleration framework that addresses this challenge by extracting and using generation-native compatibility evidence directly from the model's evolving internal states. Specifically, *Carnator* performs a lightweight early probe to construct an Early Signature from internal diffusion states, assessing reuse validity through risk-aware compatibility decisions. The same evidence characterizes reuse scope by localizing target-specific computation and guiding joint reuse of historical latent trajectories and sparse attention connectivity. Experiments on diverse text-to-video requests show that *Carnator* achieves up to 2.08$\times$ end-to-end acceleration with limited quality degradation.

1 INTRODUCTION

Video diffusion transformers have rapidly advanced in generation fidelity, temporal coherence, and controllability, fueling the growing popularity and adoption of AI-generated video across content creation, entertainment, and online media (Wan et al., 2025; Kong et al., 2024; Yang et al., 2025). However, video generation remains computationally intensive as iterative denoising repeatedly processes long spatiotemporal token sequences through large Transformer backbones. This results in substantial generation latency and prolonged GPU occupation, limiting both interactive use and serving throughput. For instance, in batch-one video generation, where lower latency improves throughput, a 3.2$\times$ speedup can reduce serving cost by about 69% at scale, highlighting the practical value of efficient inference (Shahsavan et al., 2026).

To accelerate video generation, prior work has primarily explored a broad range of intra-request acceleration techniques. These approaches exploit computational redundancy, including few-step distillation, feature caching (Nie et al., 2026; Lyu et al., 2025), token-wise computation reduction, and sparse attention (Zou et al., 2025; Chen et al., 2026). While effective, their acceleration gains are inherently bounded by the redundancy available within a single request. At serving scale, however, video generation systems operate under continuous workloads consisting of requests that often share objects, scenes, and structural patterns. This workload pattern reveals a new opportunity for inference acceleration by exploiting computational redundancy across different generation requests.

Recent cross-request methods have demonstrated that historical computation can accelerate related generation requests. However, effective reuse requires two coupled decisions, *reuse validity*, which determines whether a historical cache can be reused, and *reuse scope*, which determines where target-specific computation must be preserved once reuse is deemed valid. Reuse validity cannot be reliably inferred from request semantics alone, as semantically similar requests may develop substantially different visual structures during generation, as illustrated in Fig. **??**. Existing text-

to-video (T2V) approaches have primarily exploited intermediate latent states or region-selective recomputation, while recent sparse-attention connectivity reuse has been demonstrated mainly in image-to-video (I2V) settings with stronger visual conditioning. Reuse scope therefore remains unclear in T2V, since reusable computation may span both latent trajectories and attention connectivity. Even for a compatible history, reusable computation may vary across spatial regions and attention interactions. Overly aggressive reuse may degrade generation quality, whereas conservative reuse leaves substantial acceleration opportunities unrealized.

These challenges suggest looking beyond request semantics and into the generation process itself. Early diffusion states already reveal request-specific differences in object appearance, geometry, and spatial structure (Wen et al., 2024). Cross-attention provides noun-conditioned localization, self-attention expands this evidence over the spatial support of generated objects, and the corresponding latent states encode emerging geometric and appearance information. We refer to these request-specific internal signals as *generation-native compatibility evidence*, which better captures reuse validity than prompt semantics alone by directly reflecting the evolving target generation. The same evidence also reveals object regions and attention interaction structure, indicating where target-specific computation must be preserved and where historical computation can remain reusable. Together, these observations suggest that generation-native compatibility evidence can support both reuse validity and reuse scope within a unified cross-request acceleration framework.

Based on the observations, we present *Carnator*, a generation-native compatibility-guided framework for cross-request video generation acceleration. To assess reuse validity, *Carnator* first performs text-level recall to retrieve semantically related historical requests and then applies a lightweight early probe to construct an Early Signature, a compact summary of localized geometry and appearance in the target's early generation state. The Early Signature is compared with recalled histories under risk compatibility thresholds to select caches whose computation remains suitable for reuse. Once a cache is accepted, the same evidence is carried forward to determine reuse scope by identifying the target-specific computation that must be preserved. For accepted caches, it guides *Joint Latent-Attention Reuse*, which combines reuse of historical latent states and sparse attention connectivity while preserving target-specific computation. Extensive experiments across multiple video diffusion models demonstrate that *Carnator* consistently reduces end-to-end (E2E) inference latency while maintaining competitive generation quality.

Our main contributions are summarized as follows:

- *Carnator* introduces a generation-native compatibility assessment component for reuse validity. It performs a lightweight early probe to construct an Early Signature directly from internal diffusion states and uses risk-aware compatibility decisions to identify compatible historical computation beyond coarse semantic similarity.
- *Carnator* further develops an evidence-guided joint reuse component for reuse scope. The same generation-native evidence localizes target-specific computation and guides joint reuse of historical latent trajectories and sparse attention connectivity, reducing redundant token updates and attention interactions while preserving necessary target updates.
- We conduct extensive experiments on VidProM across three text-to-video diffusion backbones to evaluate both reuse validity and inference acceleration. *Carnator* improves cache selection reliability, reduces unsafe reuse, and achieves up to 2.08$\times$ end-to-end acceleration while maintaining competitive generation quality.

2 Related Work

Efficient diffusion inference. Diffusion models require repeated neural network evaluation over many denoising steps, and video generation further increases the cost through long spatiotemporal token sequences. Existing acceleration methods reduce this cost through step reduction, quantization, token selection, feature caching, sparse attention, and adaptive computation. Recent approaches also coordinate multiple acceleration mechanisms to reduce computation across timesteps, tokens, and attention connections (Liu et al., 2026c; Zhao et al., 2026). Most of these techniques optimize a single generation process and do not exploit reusable computation from previously completed requests.

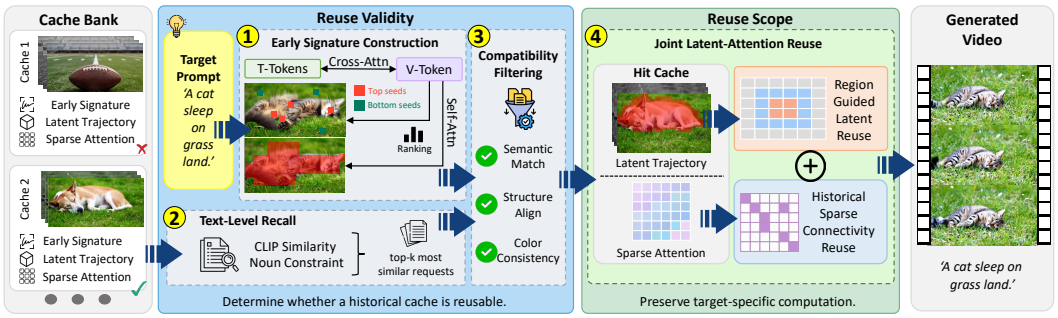


Figure 1: Overview of *Carnator*. Historical requests are first retrieved through text-level recall and filtered using generation-native Early Signatures. For an accepted cache, *Carnator* jointly reuses historical latent trajectories and sparse attention connectivity while preserving target-specific computation.

Intra-request acceleration. Intra-request methods exploit redundancy within one denoising trajectory. Feature caching reuses intermediate representations across nearby timesteps, while token-wise methods selectively refresh tokens according to their sensitivity to caching errors (Zou et al., 2025; Liu et al., 2025). Sparse attention methods reduce quadratic attention cost by identifying structured patterns across timesteps, layers, and heads, then executing only selected query-key interactions with efficient kernels (Chen et al., 2026). Diffusion attention has also been used to recover semantic regions from cross-attention and self-attention maps (Tian et al., 2024). These methods provide important foundations for selective computation, but their reuse decisions are derived from states within the current trajectory. *Carnator* instead considers whether latent states, intermediate features, token regions, and attention patterns from a different generation trajectory remain compatible with the target request.

Inter-request acceleration. Inter-request reuse has recently emerged as a new direction for diffusion serving. Early work retrieves similar T2V requests and reuses intermediate noise states to skip initial denoising computation (Agarwal et al., 2024). Subsequent methods extend this idea to T2V generation through entity-level retrieval, shared semantic components, staged latent reuse, and region-selective recomputation (Liu et al., 2026b). CHAI (Cherian et al., 2026) instantiates cross-inference cache attention on OpenSora 1.2 by modifying its STDiT attention blocks, while its portability to other video diffusion architectures remains unexplored. Historical sparse attention patterns can also be transferred across related I2V requests to reduce online mask construction (Liu et al., 2026a). *Carnator* uses generation-native compatibility evidence to address reuse validity and reuse scope in inter-request acceleration. It constructs an Early Signature for compatibility filtering according to the target generation state, and exploits accepted caches through joint region-guided latent reuse and historical sparse-connectivity reuse.

3 Method

3.1 Overview and Problem Setup

Let q denote a target video generation request and $\mathcal{B} = \{\mathcal{C}_1, \ldots, \mathcal{C}_M\}$ a cache bank constructed from completed historical requests. For each target request, *Carnator* addresses two coupled decisions, namely reuse validity, which determines whether a historical cache can be reused, and reuse scope, which determines where target-specific computation must be preserved within joint latent-attention reuse once a compatible cache is identified.

To assess reuse validity, *Carnator* first performs text-level recall to identify candidate histories with compatible noun compositions and then runs a lightweight target probe to construct an Early Signature S_q from the model's internal generation states. The target signature is compared with the stored signatures of recalled caches under calibrated compatibility thresholds. If no candidate is

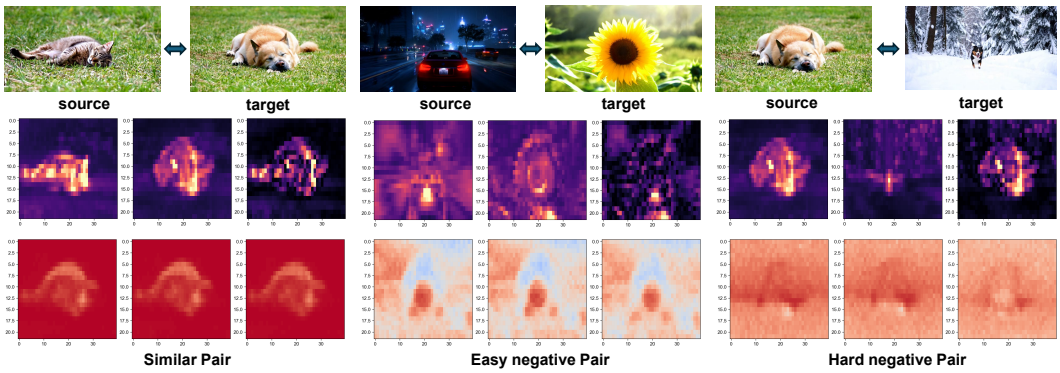


Figure 2: Early generation states provide informative signals for cross-request compatibility. Similar request pairs exhibit stronger alignment in early denoising representations, while easy and hard negative pairs show increasingly distinct prompt-conditioned residuals and latent dynamics.

compatible, the target proceeds with dense generation. Otherwise, an accepted cache $\mathcal{C}_{j^\star}$ is selected for subsequent reuse.

For an accepted cache, *Carnator* carries the same generation-native compatibility evidence forward to determine reuse scope. The localized evidence identifies where target-specific computation must be preserved within joint latent-attention reuse. The same generation-native evidence therefore connects the two stages, with the Early Signature supporting reuse-validity assessment and its localized evidence guiding reuse scope during subsequent inference. The overall workflow is illustrated in Fig. 1. Sec. 3.2 describes reuse validity, while Sec. 3.3 describes reuse scope.

3.2 Reuse Validity with Generation-Native Evidence

Early Signature Construction *Carnator* requires generation-native evidence before committing to cache reuse. For a request r with prompt y_r, *Carnator* performs a lightweight early probe and constructs an Early Signature S_r from the partially formed generation state. Each noun entity is localized from the model's internal cross- and self-attention states and summarized using compact descriptors for subsequent compatibility filtering. Early diffusion states already provide informative evidence for reuse-validity assessment, as illustrated in Fig. 2.

For each noun entity $o \in \mathcal{N}(y_r)$, *Carnator* first extracts noun-conditioned seed responses from cross-attention, following prior use of cross- and self-attention for semantic localization (Liu et al., 2024; Qi et al., 2023). For visual token $i \in \Omega$, $A^{\text{cross}}_{l,h}[i,o]$ denotes its cross-attention weight to noun o at Transformer layer l and head h. These responses are aggregated as $C^o_i = \text{Agg}_{l,h} A^{\text{cross}}_{l,h}[i,o]$, with high- and low-response tokens forming the noun and noise seed sets $\mathcal{I}^+_o$ and $\mathcal{I}^-_o$, respectively. The seeds are then expanded through self-attention using $S^o_i = \text{Agg}_h \left[\max_{u\in\mathcal{I}^+_o} A^{\text{self}}_h[i,u] - \max_{u\in\mathcal{I}^-_o} A^{\text{self}}_h[i,u]\right]$. Fusing C^o_i and S^o_i and thresholding the resulting object-support score yields the Evidence Region $R^{\text{evidence}}_{r,o} \subseteq \Omega$.

Carnator converts the Evidence Region into a binary spatial indicator $m_{r,o} \in \{0,1\}^{H\times W}$. Its normalized area and spatial centroid are $A_{r,o} = \sum_{h,w} m_{r,o}(h,w)/(HW)$ and $\boldsymbol{\mu}_{r,o}$, respectively. Appearance is represented by the mean CIELAB color $\mathbf{c}^{\text{Lab}}_{r,o} \in \mathbb{R}^3$ over the corresponding decoded region. The resulting descriptor is $s_{r,o} = (m_{r,o}, A_{r,o}, \boldsymbol{\mu}_{r,o}, \mathbf{c}^{\text{Lab}}_{r,o})$.

We collect these noun-level descriptors into the Early Signature $S_r = \{s_{r,o} \mid o \in \mathcal{N}(y_r)\}$, which is stored with the reusable latent states and sparse connectivity of each completed historical request.

Compatibility Filtering Given a target request q and cache bank $\mathcal{B}$, *Carnator* first performs text-level recall and then compares Early Signatures for compatibility filtering.

Carnator retrieves the top-K historical requests with the highest CLIP text similarity (Radford et al., 2021), forming the coarse candidate set $\widetilde{\mathcal{L}}_q$. Let $\mathcal{N}(y)$ denote the noun set extracted from prompt y. A candidate $\mathcal{C}_j \in \widetilde{\mathcal{L}}_q$ is retained only when $|\mathcal{N}(y_q) \setminus \mathcal{N}(y_j)| = |\mathcal{N}(y_j) \setminus \mathcal{N}(y_q)| \leq 1$. For each retained candidate, $\mathcal{O}_{\text{same}} = \mathcal{N}(y_q) \cap \mathcal{N}(y_j)$ denotes the shared nouns, while o_q and o_j denote the target-only and historical-only nouns when one substitution exists. The retained candidates preserve their CLIP-similarity order.

The target constructs its Early Signature S_q once and compares it with the stored signature S_j of each retained candidate $\mathcal{C}_j$. For the shared nouns, *Carnator* measures the maximum appearance discrepancy $D_{\text{context}} = \max_{o \in \mathcal{O}_{\text{same}}} \Delta E_{00} \left(\mathbf{c}^{\text{Lab}}_{q,o}, \mathbf{c}^{\text{Lab}}_{j,o} \right)$, where $D_{\text{context}} = 0$ when $\mathcal{O}_{\text{same}}$ is empty. When one noun substitution exists, *Carnator* further compares the size, shape, and appearance of o_q and o_j as

$$d_{\text{size}} = \left| \log \frac{A_{q,o_q}}{A_{j,o_j}} \right|, \qquad d_{\text{shape}} = 1 - \frac{2|\widetilde{m}_{q,o_q} \cap \widetilde{m}_{j,o_j}|}{|\widetilde{m}_{q,o_q}| + |\widetilde{m}_{j,o_j}|}, \qquad d_{\text{color}} = \Delta E_{00} \left(\mathbf{c}^{\text{Lab}}_{q,o_q}, \mathbf{c}^{\text{Lab}}_{j,o_j} \right).$$

Here, $\widetilde{m}_{q,o_q}$ and $\widetilde{m}_{j,o_j}$ denote centroid-aligned and scale-normalized object indicators.

Given thresholds $\boldsymbol{\tau} = (\tau_{\text{context}}, \tau_{\text{size}}, \tau_{\text{shape}}, \tau_{\text{color}})$, the compatibility decision is

$$g_{\boldsymbol{\tau}}(q, \mathcal{C}_j) = \mathbb{I}[D_{\text{context}} \leq \tau_{\text{context}}] \mathbb{I}[d_{\text{size}} \leq \tau_{\text{size}}] \mathbb{I}[d_{\text{shape}} \leq \tau_{\text{shape}}] \mathbb{I}[d_{\text{color}} \leq \tau_{\text{color}}].$$

For identical noun sets, only the shared-noun context criterion is applied. *Carnator* selects the first compatible candidate in CLIP order and falls back to dense generation if none satisfies the rule.

Risk-Calibrated Compatibility Decisions *Carnator* calibrates $\boldsymbol{\tau}$ from observed reuse outcomes, labeling each request–cache pair as $y_i = \mathbb{I}[\Delta Q_i^{(m)} \leq \epsilon_m, \ \forall m \in \mathcal{M}]$, where $y_i = 1$ denotes safe reuse. On the threshold-selection set, we choose $\boldsymbol{\tau}^\star = \arg\max_{\boldsymbol{\tau}} C(\boldsymbol{\tau})$ s.t. $R(\boldsymbol{\tau}) \leq \delta$, with $C(\boldsymbol{\tau}) = \frac{1}{N} \sum_i g_{\boldsymbol{\tau}}(i)$, $R(\boldsymbol{\tau}) = \frac{\sum_i g_{\boldsymbol{\tau}}(i)(1-y_i)}{\sum_i g_{\boldsymbol{\tau}}(i)}$. The selected $\boldsymbol{\tau}^\star$ is then evaluated on a disjoint certification set and accepted only when the one-sided Clopper–Pearson upper bound satisfies $U_\alpha(u, n) \leq \delta$ for n accepted pairs with u unsafe cases, at confidence level $1 - \alpha$.

3.3 Reuse Scope with Joint Latent-Attention Reuse

Reuse validity determines whether a historical cache can be used. Once accepted, *Carnator* uses the same generation-native compatibility evidence to determine where target-specific computation must be preserved within a joint latent-attention reuse path. *Carnator* jointly exploits cached latent states and historical sparse attention connectivity, while restricting target-specific computation to the evidence-derived regions and their associated attention domain. The target generation resumes from a cached intermediate state and continues denoising under the target prompt, preserving compatible historical latent states outside target-specific regions while reusing historical sparse connectivity to reduce redundant query–key interactions. Fig. 3 illustrates how the two forms of reuse are integrated across the DiT computation path.

Region-Guided Latent Reuse For latent reuse, *Carnator* uses the object regions obtained from the accepted cache to localize target-specific computation. The Evidence Region $\mathcal{R}_{\text{evidence}}$ is spatially expanded into an Active Region $\mathcal{R}_{\text{active}}$ and a larger Context Region $\mathcal{R}_{\text{context}}$, satisfying $\mathcal{R}_{\text{evidence}} \subseteq \mathcal{R}_{\text{active}} \subseteq \mathcal{R}_{\text{context}}$. Tokens within $\mathcal{R}_{\text{active}}$ are updated according to the target request, while the remaining latent states are inherited from the historical trajectory. Within selected middle Transformer layers, active tokens are recomputed using contextual keys and values from $\mathcal{R}_{\text{context}}$, whereas tokens outside $\mathcal{R}_{\text{active}}$ preserve their cached computation.

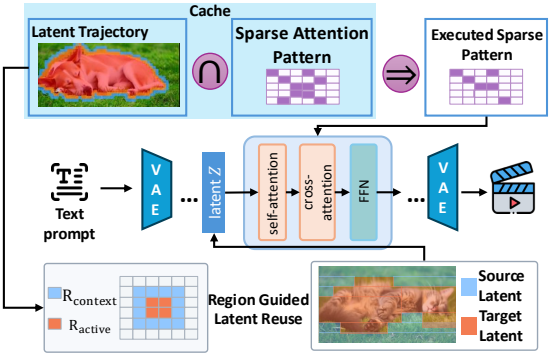


Figure 3: Joint latent-attention reuse in *Carnator*.

We focus region-restricted updates on intermediate layers, motivated by prior observations that intermediate diffusion representations contain richer visual-semantic information (Pnvr et al., 2023). For each subsequent denoising step, the resulting latent can be expressed as

$$\mathbf{z}_{k+1} = \mathbf{1}_{\mathcal{R}_{\text{active}}} \odot \widehat{\mathbf{z}}^{q}_{k+1} + (1 - \mathbf{1}_{\mathcal{R}_{\text{active}}}) \odot \mathbf{z}^{j^\star}_{k+1},$$

where $\widehat{\mathbf{z}}^{q}_{k+1}$ denotes the target-updated latent and $\mathbf{z}^{j^\star}_{k+1}$ denotes the corresponding historical latent.

Historical Sparse-Connectivity Reuse In parallel, *Carnator* reuses the sparse attention structure recorded during the historical generation. For each denoising step, Transformer layer, and attention head, the historical request stores the blockwise connectivity that captures its dominant query–key interactions. During target inference, *Carnator* recomputes the target $\mathbf{Q}$, $\mathbf{K}$, and $\mathbf{V}$ representations but reuses the historical sparse block structure to determine which query–key interactions are evaluated. Thus, the transferred computation describes *where* attention is performed rather than reusing historical attention values themselves. This allows the target request to preserve its own representations while avoiding the dense attention computation associated with structurally redundant interactions.

Joint Reuse in Intermediate Layers The two forms of reuse are combined in the middle Transformer layers. In these layers, historical sparse connectivity is further constrained by the target-specific computation domain defined by $\mathcal{R}_{\text{active}}$ and $\mathcal{R}_{\text{context}}$. Only historical connectivity associated with active queries and their valid contextual tokens is retained, so attention reuse is restricted to the portion of the historical structure that remains relevant to the target generation. Other Transformer layers directly reuse the historical sparse block structure without this additional region constraint. In this way, *Carnator* combines region-guided latent reuse with historical sparse-connectivity reuse, reducing computation along both the token and attention dimensions while preserving target-specific semantic updates.

4 Experiments

4.1 Experimental Setup

Models and Hardware. We evaluate all methods on three video diffusion models, Wan2.2-TI2V-5B, Wan2.1-T2V-1.3B (Wan et al., 2025), and LTX-Video-13B (HaCohen et al., 2024). Wan2.2-TI2V-5B serves as our primary evaluation model and generates 121 frames at 1280×704 resolution and 24 FPS using 50-step UniPC sampling. Within each model, all methods use the same model-specific generation configuration, random-seed protocol, precision, and decoding pipeline. Experiments are conducted on NVIDIA H100 GPUs.

Workload and Cache Settings. We construct the serving workload from VidProM (**?**), with 1,000 historical requests as a fixed warm cache and a disjoint 300-request test stream. Within each model, all cross-request methods use the same cache and request order. Cache misses fall back to dense generation, and all online costs, including retrieval, early probing, signature construction, compatibility filtering, reuse, fallback, and decoding, are included in end-to-end latency for cache-hit requests.

Baselines. We compare *Carnator* with dense inference, NIRVANA (Agarwal et al., 2024), Chorus (Liu et al., 2026b), and Chorus II (Liu et al., 2026a). NIRVANA is originally designed for text-to-image (T2I) generation, and we adapt its cross-request latent reuse to T2V. Chorus is evaluated in its original T2V setting, while Chorus II, originally developed for I2V, is adapted to T2V by transferring its historical sparse-attention reuse mechanism. All methods share the same backbone, workload, and cache setting. CHAI (Cherian et al., 2026) is evaluated separately in Appendix C due to its OpenSora 1.2-specific STDiT design.

Metrics. We evaluate both generation quality and inference efficiency. Generation quality is measured by VBench-Q (Huang et al., 2024) and CLIP-T (Radford et al., 2021), capturing general video quality and text-condition alignment, respectively. Efficiency is measured by end-to-end latency, DiT latency and speedup over dense inference.

Table 1: End-to-end performance on the VidProM workload across three video diffusion models. Within each model, all methods process the same request sequence under the same generation configuration. For cross-request methods, cache misses fall back to dense generation.

Method	CLIP-T ↑	VBench-Q ↑	E2E Latency (s) ↓	DiT Latency (s) ↓	E2E Speedup ↑	DiT Speedup ↑	Hit Rate ↑
Wan2.2-TI2V-5B							
Dense	0.3372	0.6331	293.88	265.23	1.00×	1.00×	N/A
NIRVANA-VID	0.2855	0.6487	256.09	204.21	1.15×	1.30×	80.67%
Chorus	0.2917	0.6407	264.47	175.48	1.11×	1.51×	80.67%
Chorus II	0.2893	0.6417	222.44	201.58	1.32×	1.32×	80.67%
CARNATOR	0.2740	0.6401	204.30	159.65	1.44×	1.66×	72.67%
Wan2.1-T2V-1.3B							
Dense	0.2984	0.6379	1138.6	974.2	1.00×	1.00×	N/A
NIRVANA-VID	0.2934	0.6169	1035.9	884.5	1.10×	1.10×	80.67%
Chorus	0.2899	0.6019	719.8	582.4	1.58×	1.67×	80.67%
Chorus II	0.2869	0.6461	577.1	556.9	1.97×	1.75×	80.67%
CARNATOR	0.2734	0.6356	547.2	496.0	2.08×	1.96×	77.67%
LTX-Video-13B							
Dense	0.2608	0.5896	81.42	79.37	1.00×	1.00×	N/A
NIRVANA-VID	0.2581	0.6099	72.82	70.78	1.12×	1.12×	80.67%
Chorus	0.2566	0.6018	136.24	70.78	0.60×	1.12×	80.67%
Chorus II	0.2771	0.6328	77.69	75.26	1.05×	1.05×	80.67%
CARNATOR	0.2577	0.6022	68.15	62.53	1.19×	1.27×	79.33%

4.2 Main Results

Table 1 summarizes the end-to-end performance of all methods on the VidProM workload. *Carnator* achieves the lowest end-to-end and DiT latencies among cache-hit requests, reducing them to 204.30 s and 159.65 s, respectively, corresponding to 1.44× end-to-end and 1.66× DiT speedups over dense inference. It also provides the highest end-to-end acceleration among the evaluated cross-request baselines, outperforming NIRVANA-VID, Chorus, and Chorus II. These results demonstrate the effectiveness of exploiting reusable computation across both latent trajectories and attention connectivity.

The efficiency gains are particularly pronounced within the DiT backbone. Despite accepting fewer historical caches, Carnator achieves the lowest latency, indicating that its joint latent-attention reuse extracts larger computational savings from each accepted cache. Region-guided latent reuse reduces redundant token updates, while historical sparse-connectivity reuse avoids unnecessary query–key interactions during attention computation. Their joint use therefore increases the amount of computation that can be reused from each compatible historical request.

In terms of generation quality, *Carnator* achieves a VBench-Q score of 0.6401, comparable to dense inference and the evaluated acceleration baselines, together with a CLIP-T score of 0.2740. The VBench-Q results indicate that the proposed reuse mechanism preserves the major visual-quality dimensions captured by VBench while substantially reducing inference latency. Across the evaluated methods, the results reveal a favorable quality–efficiency trade-off for cross-request computation reuse.

Overall, the results highlight two complementary factors in cross-request acceleration. Reuse validity determines whether historical computation can be reused, while reuse scope determines where target-specific computation must be preserved within joint latent-attention reuse. Together, these two components enable *Carnator* to translate cross-request redundancy into substantial end-to-end inference savings.

4.3 Ablation Studies

Generation-Level Compatibility. Table 2 evaluates the effect of incorporating the Early Signature into cross-request cache retrieval. Compared with text-level retrieval alone, generation-level filtering reduces the cache hit rate from 77.33% to 72.67%, while improving CLIP-T from 0.2625 to 0.2740 and maintaining a comparable VBench-Q score of 0.6401. This indicates that the Early Signature filters out a small fraction of textually similar but generation-incompatible caches, leading to better text–video alignment without sacrificing overall visual quality. Meanwhile, the end-to-end

latency remains essentially unchanged, showing that the additional compatibility filtering introduces little overhead relative to the computation saved by cache reuse.

On a disjoint certification set, the calibrated compatibility rule accepts 50 of 70 request-cache pairs with no unsafe cases. The resulting one-sided Clopper-Pearson upper bound is 4.50 percent, satisfying the prescribed 5 percent risk constraint. Model-specific compatibility calibration follows the same protocol, with details provided in Appendix D.

Table 2: Ablation of reuse-validity assessment on the VidProM workload. Speedup is calculated with respect to dense generation.

Retrieval / Gate	Hit Rate	CLIP-T ↑	VBench-Q ↑	E2E Latency (s) ↓	Speedup ↑
Text-Level Retrieval	77.33%	0.2625	0.6397	203.6	1.44×
+ Early Signature	72.67%	0.2740	0.6401	204.30	1.44×

Joint Latent-Attention Reuse. Table 3 evaluates the individual contributions of latent reuse and attention reuse on a randomly sampled fixed subset of 10 requests. Latent reuse reduces redundant token updates and lowers end-to-end latency to 206.06 s, while attention reuse decreases query–key computation and achieves 160.91 s. Combining the two mechanisms achieves the lowest latency of 155.85 s and the highest end-to-end speedup of 1.71×, confirming that the two forms of reuse provide complementary acceleration benefits. Across all variants, CLIP-T and VBench-Q remain within a narrow range, indicating that the additional acceleration from joint reuse does not substantially compromise generation quality.

Table 3: Ablation of latent and attention reuse in *CARNATOR*.

Variant	Latency (s) ↓	Speedup ↑	CLIP-T ↑	VBench-Q ↑
Full Computation	266.26	1.00×	0.3391	0.6432
Latent Reuse Only	206.06	1.29×	0.3237	0.6436
Attention Reuse Only	160.91	1.65×	0.3291	0.6291
Joint Latent-Attention Reuse	155.85	1.71×	0.3281	0.6433

5 Conclusion

In this work, we presented *Carnator*, a cross-request acceleration framework for text-to-video diffusion models that uses generation-native compatibility evidence to jointly address reuse validity and reuse scope. *Carnator* constructs an Early Signature from early diffusion states to identify reusable historical caches beyond prompt similarity, and applies risk-aware compatibility decisions. For accepted caches, it jointly exploits redundancy in latent trajectories and attention connectivity through region-guided latent reuse and historical sparse-connectivity reuse. Experiments demonstrate that *Carnator* reduces video diffusion inference cost while maintaining competitive generation quality across diverse generation requests. More broadly, our results show that cross-request redundancy provides a complementary dimension for accelerating video generation beyond conventional optimization within individual denoising trajectories.

References

Shubham Agarwal, Subrata Mitra, Sarthak Chakraborty, Srikrishna Karanam, Koyel Mukherjee, and Shiv Kumar Saini. Approximate caching for efficiently serving text-to-image diffusion models. In *21st USENIX Symposium on Networked Systems Design and Implementation (NSDI 24)*, pp. 1173–1189, 2024.

Pengtao Chen, Xianfang Zeng, Maosen Zhao, Mingzhu Shen, Wei Cheng, Gang Yu, and Tao Chen. Sparse-vdit: Unleashing the power of sparse attention to accelerate video diffusion transformers. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 40, pp. 2957–2965, 2026.

Joel Mathew Cherian, Ashutosh Muralidhara Bharadwaj, Vima Gupta, and Anand Padmanabha Iyer. Chai: Cache attention inference for text2video. *arXiv preprint arXiv:2602.16132*, 2026.

Yoav HaCohen, Nisan Chiprut, Benny Brazowski, Daniel Shalem, Dudu Moshe, Eitan Richardson, Eran Levin, Guy Shiran, Nir Zabari, Ori Gordon, et al. Ltx-video: Realtime video latent diffusion. *arXiv preprint arXiv:2501.00103*, 2024.

Ziqi Huang, Yinan He, Jiashuo Yu, Fan Zhang, Chenyang Si, Yuming Jiang, Yuanhan Zhang, Tianxing Wu, Qingyang Jin, Nattapol Chanpaisit, et al. Vbench: Comprehensive benchmark suite for video generative models. In *2024 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 21807–21818. IEEE, 2024.

Weijie Kong, Qi Tian, Zijian Zhang, Rox Min, Zuozhuo Dai, Jin Zhou, Jiangfeng Xiong, Xin Li, Bo Wu, Jianwei Zhang, et al. Hunyuanvideo: A systematic framework for large video generative models. *arXiv preprint arXiv:2412.03603*, 2024.

Bingyan Liu, Chengyu Wang, Tingfeng Cao, Kui Jia, and Jun Huang. Towards understanding cross and self-attention in stable diffusion for text-guided image editing. In *2024 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 7817–7826. IEEE, 2024.

Hao Liu, Chenghuan Huang, Xing Cai, Chen Li, Ziyang Ma, Jing Lyu, Nong Xiao, and Jiangsu Du. Chorus ii: Cross-request sparsity reuse for efficient image-to-video generation. *arXiv preprint arXiv:2606.25040*, 2026a.

Hao Liu, Ye Huang, Chenghuan Huang, Zhenyi Zheng, Jiangsu Du, Ziyang Ma, Jing Lyu, and Yutong Lu. Beyond few-step inference: Accelerating video diffusion transformer model serving with inter-request caching reuse. *arXiv preprint arXiv:2604.04451*, 2026b.

Haosong Liu, Yuge Cheng, Wenxuan Miao, Zihan Liu, Aiyue Chen, Jing Lin, Yiwu Yao, Chen Chen, Jingwen Leng, Minyi Guo, and Yu Feng. Astraea: A token-wise acceleration framework for video diffusion transformers. In C. Vondrick, B. Hariharan, C. Raffel, L. Pinto, D. Yang, and A. Faust (eds.), *International Conference on Learning Representations*, volume 2026, pp. 90226–90250, 2026c. URL https://proceedings.iclr.cc/paper_files/paper/2026/file/921ac785fa9edc73cacaf2664f43d234-Paper-Conference.pdf.

Jiacheng Liu, Chang Zou, Yuanhuiyi Lyu, Fei Ren, Shaobo Wang, Kaixin Li, and Linfeng Zhang. Speca: Accelerating diffusion transformers with speculative feature caching. In *Proceedings of the 33rd ACM International Conference on Multimedia*, pp. 10024–10033, 2025.

Zhengyao Lyu, Chenyang Si, Junhao Song, Zhenyu Yang, Yu Qiao, Ziwei Liu, and Kwan-Yee K Wong. Fastercache: Training-free video diffusion model acceleration with high quality. In *International Conference on Learning Representations*, volume 2025, pp. 33132–33156, 2025.

Weili Nie, Julius Berner, Nanye Ma, Chao Liu, Saining Xie, and Arash Vahdat. Transition matching distillation for fast video generation. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 4645–4655, June 2026.

Koutilya Pnvr, Bharat Singh, Pallabi Ghosh, Behjat Siddiquie, and David Jacobs. Ld-znet: A latent diffusion approach for text-based image segmentation. In *Proceedings of the IEEE/CVF international conference on computer vision*, pp. 4157–4168, 2023.

Chenyang Qi, Xiaodong Cun, Yong Zhang, Chenyang Lei, Xintao Wang, Ying Shan, and Qifeng Chen. Fatezero: Fusing attentions for zero-shot text-based video editing. In *2023 IEEE/CVF International Conference on Computer Vision (ICCV)*, pp. 15886–15896. IEEE, 2023.

Alec Radford, Jong Wook Kim, Chris Hallacy, Aditya Ramesh, Gabriel Goh, Sandhini Agarwal, Girish Sastry, Amanda Askell, Pamela Mishkin, Jack Clark, et al. Learning transferable visual models from natural language supervision. In *International conference on machine learning*, pp. 8748–8763. PmLR, 2021.

Faraz Shahsavan, Yikai Zhu, Pankaj Gupta, and Philip Kiely. Wan 2.2 video generation in less than 60 seconds. https://www.baseten.co/blog/wan-2-2-video-generation-in-less-than-60-seconds/, August 2026. Baseten Blog, accessed September 14, 2026.

Junjiao Tian, Lavisha Aggarwal, Andrea Colaco, Zsolt Kira, and Mar Gonzalez-Franco. Diffuse, attend, and segment: Unsupervised zero-shot segmentation using stable diffusion. In *2024 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 3554–3563. IEEE, 2024.

Team Wan, Ang Wang, Baole Ai, Bin Wen, Chaojie Mao, Chen-Wei Xie, Di Chen, Feiwu Yu, Haiming Zhao, Jianxiao Yang, et al. Wan: Open and advanced large-scale video generative models. *arXiv preprint arXiv:2503.20314*, 2025.

Yuxin Wen, Yuchen Liu, Chen Chen, and Lingjuan Lyu. Detecting, explaining, and mitigating memorization in diffusion models. In *The Twelfth International Conference on Learning Representations*, 2024.

Zhuoyi Yang, Jiayan Teng, Wendi Zheng, Ming Ding, Shiyu Huang, Jiazheng Xu, Yuanming Yang, Wenyi Hong, Xiaohan Zhang, Guanyu Feng, et al. Cogvideox: Text-to-video diffusion models with an expert transformer. In *International Conference on Learning Representations*, volume 2025, pp. 83048–83077, 2025.

Wangbo Zhao, Yizeng Han, Zhiwei Tang, Jiasheng Tang, Pengfei Zhou, Kai Wang, Bohan Zhuang, Zhangyang Wang, Fan Wang, and Yang You. Rapid 3: Tri-level reinforced acceleration policies for diffusion transformer. In *International Conference on Learning Representations*, volume 2026, pp. 149423–149446, 2026.

Chang Zou, Xuyang Liu, Ting Liu, Siteng Huang, and Linfeng Zhang. Accelerating diffusion transformers with token-wise feature caching. In Y. Yue, A. Garg, N. Peng, F. Sha, and R. Yu (eds.), *International Conference on Learning Representations*, volume 2025, pp. 75467–75488, 2025. URL https://proceedings.iclr.cc/paper_files/paper/2025/file/bbe024e0517fe12ac3a8d388b19ff9fe-Paper-Conference.pdf.

A Additional Analysis

A.1 Early-State Generation-Native Compatibility Evidence

Fig. 4 provides additional analysis of early latent dynamics. Similar request pairs exhibit consistently smaller source–target discrepancies than incompatible pairs, providing further evidence that early diffusion states contain useful generation-native compatibility evidence for reuse-validity assessment.

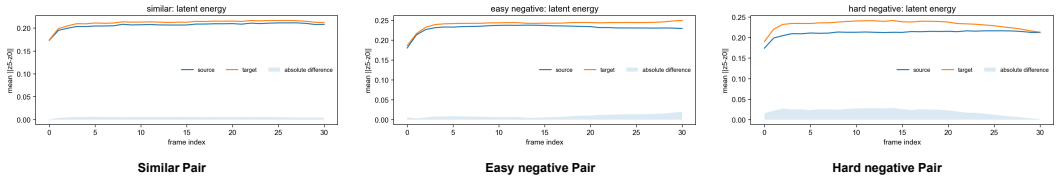


Figure 4: Latent-state discrepancy for similar, easy-negative, and hard-negative request pairs across video frames.

A.2 Cross-Request Attention Similarity

Fig. 5 visualizes the blockwise attention structure of two semantically related requests. Despite the object substitution, the two requests exhibit highly similar attention mass distributions and sparse connectivity patterns, motivating Historical Sparse-Connectivity Reuse.

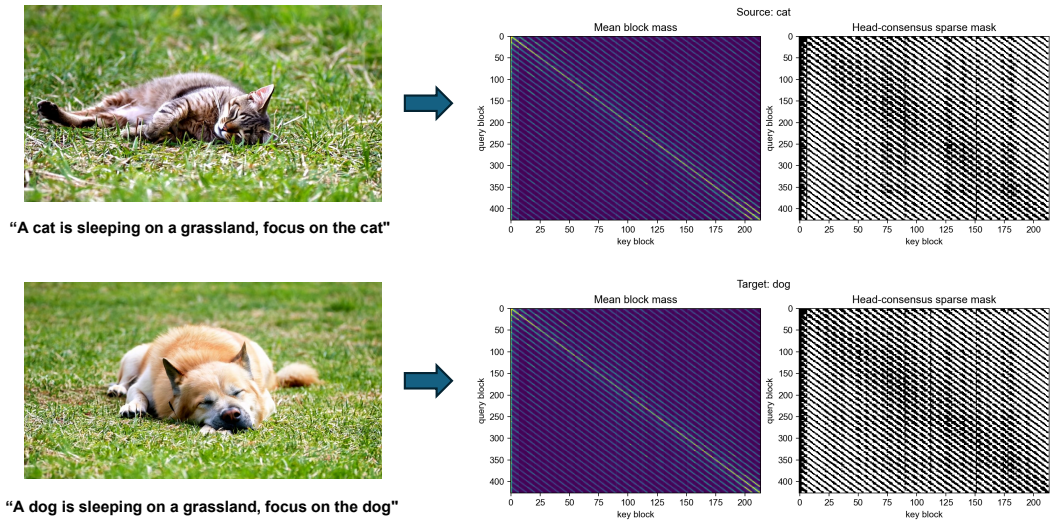


Figure 5: Blockwise attention mass and sparse connectivity for two related text-to-video requests. Similar requests preserve substantial structural similarity in their attention patterns despite changes in object identity.

B Additional Qualitative Results

This section provides additional qualitative results of *Carnator* on diverse text-to-video generation requests. The examples illustrate that *Carnator* preserves target-specific content and overall visual quality while reusing computation from compatible historical requests.

Place Holder

Figure 6: Qualitative result of *Carnator*

C Cross-Architecture Portability of CHAI

CHAI (Cherian et al., 2026) designs Cache Attention specifically for the STDiT architecture of OpenSora 1.2. We further examine its cross-architecture portability by directly adapting the same cache-reuse mechanism to Wan2.2. As illustrated in Fig. **??**, CHAI preserves generation quality on its native OpenSora 1.2 architecture, whereas the direct Wan2.2 adaptation exhibits noticeable visual degradation. This qualitative comparison suggests that CHAI's Cache Attention mechanism requires architecture-specific adaptation when transferred to Wan2.2.

Table 4: Risk calibration and certification results. The target unsafe-reuse rate is $\delta = 5\%$ at a one-sided confidence level of 90%.

$\boldsymbol{\tau}^\star$	Coverage	Accepted n	Unsafe u	Empirical Risk	$U_\alpha(u, n)$
$(1.62,\ 2.17,\ 1.00,\ 7.58)$	71.43%	50	0	0.00%	4.50%



Figure 7: Qualitative comparison of CHAI across video diffusion architectures. From left to right, we show the OpenSora 1.2 baseline, CHAI on OpenSora 1.2, the Wan2.2 baseline, and a direct adaptation of CHAI to Wan2.2. CHAI preserves generation quality on its native OpenSora architecture, while the direct Wan2.2 adaptation exhibits noticeable visual degradation.

D Risk Calibration and Certification

We use a threshold-selection set of 30 request–cache pairs and a disjoint certification set of 70 pairs. The calibrated compatibility thresholds are

$$\boldsymbol{\tau}^\star = (\tau_{\text{context}}, \tau_{\text{size}}, \tau_{\text{shape}}, \tau_{\text{color}}) = (1.62,\ 2.17,\ 1.00,\ 7.58).$$

We set the target unsafe-reuse rate to $\delta = 5\%$ and use a one-sided confidence level of 90% ($\alpha = 0.10$). On the certification set, the calibrated compatibility rule accepts 50 of the 70 request–cache pairs, with no unsafe cases. This corresponds to a reuse coverage of 71.43% and an empirical unsafe-reuse rate of 0%. The one-sided Clopper–Pearson upper bound is $U_{0.10}(0, 50) = 4.50\%$, satisfying the prescribed risk constraint.

E Latency Accounting

E.1 End-to-end latency

E2E latency is the wall-clock time of one request: the timer opens at the start of the request and closes when the decoded video is ready. It therefore covers retrieval, text encoding and initial latent construction, the entire denoising loop (DiT forwards and scheduler updates, latent injection/fusion, and in-loop mask or index lookups), any dense probe prefix a method requires, method preprocessing that is not hoisted out (e.g. Chorus's LLM prompt-diff and segmentation), and VAE decoding of the final latent.

E.2 DiT latency

DiT latency isolates the denoiser: the CUDA-synchronized wall time of the transformer forward passes alone, summed over the executed steps. Synchronizing at both ends means the interval contains only that forward's own kernels — it never absorbs the launch tail of the preceding operation, nor the scheduler update, CFG/STG combination, latent injection or fusion, mask lookups, VAE decoding, or text encoding. Under classifier-free guidance the model runs once per branch and all branches are counted, so the metric scales with the number of forwards rather than sampler steps; where a pipeline batches its guidance branches into a single call, that call counts once. Because a method may add work before reuse, any dense probe prefix is included here too, and its steps are added to the executed-step count, so the reported DiT time always covers every forward the request actually ran. A per-module CUDA-event profiler (self-attention, cross-attention, FFN) is recorded alongside as a cross-check.

E.3 WORKED EXAMPLE: CARNATOR ON WAN2.2 TI2V-5B

Mean over the 300 hit requests of the VidProM Wan2.2 experiment. Carnator's target request runs a 5-step dense early probe for Early Signature construction and then 40 reuse steps, so 45 steps are executed and the probe's forwards count toward DiT latency.

Table 5: Latency decomposition of one Carnator target request on Wan2.2 TI2V-5B (mean over hit requests; bf16, H100 PCIe, 704×1280, 121 frames, 50-step UniPC, guidance 5.0). DiT latency covers every forward the request executes, including the 5 dense probe steps.

Component	Time (s)	Share of E2E
E2E latency	**205.18**	100 %
DiT latency (all forwards of the request)	**160.26**	78.1 %
reuse-stage DiT — 40 steps, 80 forwards at 1.66 s	133.01	64.8 %
probe-stage DiT — 5 steps, 10 forwards at 2.73 s	27.25	13.3 %
Probe stage, non-DiT (Early Signature construction & compatibility filtering)	20.26	9.9 %
Reuse stage, non-DiT (retrieval, injection, scheduler, VAE decode, video write)	24.66	12.0 %